

 Reliable INSTITUTE Unleashing Potential	MATHEMATICS REVISION PLAN-3 Target : JEE (Advanced)-2023	DPP DAILY PRACTICE PROBLEM DPP NO. # 01 (Advanced)
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Max. Time : 129 Minutes			Total Marks-166
Q. 01 to 06	Comprehension	('–1' negative marking)	[3 Marks,3 Min.][18,18]
Q. 07 to 26	Multiple choice objective	('–1' negative marking)	[4 Marks,3 Min.][80,60]
Q. 27 to 28	Match the column	(no negative marking)	[4 Marks,3 min.][8,6]
Q. 29 to 43	Numerical Type	('–1' negative marking)	[4 Marks,3 Min.][60,45]

DPP Syllabus : Vector, Complex Number

Comprehension # 1 (Q.No. 1 to 2)

Consider the line $L : \frac{x-1}{2} = \frac{y}{1} = \frac{z+1}{-2}$ and a point $A(1, 1, 1)$. Let P be the foot of perpendicular from A on L and Q be the image of point A in the line L , 'O' being the origin.

- Shortest distance of origin from the plane passing through the point A and containing the line L , is
 (A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$
- Shortest distance of line joining OP & BQ ; $B(-1, 1, 2)$ is
 (A) $\frac{1}{\sqrt{7}}$ (B) $\frac{1}{\sqrt{14}}$ (C) $\frac{2}{\sqrt{14}}$ (D) $\frac{3}{\sqrt{14}}$

Comprehension # 2 (Q.No. 3 to 4)

Let A, B, C be three sets of complex numbers as defined below.

$$A = \{z : |z+1| \leq 2 + \operatorname{Re}(z)\}, B = \{z : |z-1| \geq 1\} \text{ and } C = \left\{z : \left| \frac{z-1}{z+1} \right| \geq 1\right\}$$

- The area of region (in sq. units) bounded by $A \cap B \cap C$ is
 (A) $2\sqrt{3}$ (B) $\sqrt{3}$ (C) $4\sqrt{3}$ (D) 2
- The number of points having integral coordinates in the region $A \cap B \cap C$ is
 (A) 10 (B) 6 (C) 5 (D) 4

Comprehension # 3 (Q. No. 5 to 6)

If $\vec{x} \times \vec{y} = \vec{a}$... (i)

$\vec{y} \times \vec{z} = \vec{b}$... (ii)

$\vec{x} \cdot \vec{b} = r$... (iii)

$\vec{x} \cdot \vec{y} = 1$... (iv)

& $\vec{y} \cdot \vec{z} = 1$... (v),

where $\vec{a}, \vec{b}$ are non-zero, non-collinear vectors, then

5. Vector $\vec{x}$ is
- (A) $\frac{1}{|\vec{a} \times \vec{b}|^2} [\vec{a} \times (\vec{a} \times \vec{b})]$ (B) $\frac{r}{|\vec{a} \times \vec{b}|^2} [\vec{a} \times \vec{b} - \vec{a} \times (\vec{a} \times \vec{b})]$
- (C) $\frac{r}{|\vec{a} \times \vec{b}|^2} [\vec{a} \times \vec{b} + \vec{b} \times (\vec{a} \times \vec{b})]$ (D) None of these
6. $\vec{z}$ is
- (A) $\frac{r}{|\vec{a} \times \vec{b}|^2} [\vec{a} + \vec{b} + (\vec{a} \times \vec{b})]$ (B) $\frac{r}{|\vec{a} \times \vec{b}|^2} [\vec{a} + \vec{b} - \vec{a} \times (\vec{a} \times \vec{b})]$
- (C) $\frac{r}{|\vec{a} \times \vec{b}|^2} [\vec{a} \times \vec{b} + \vec{b} \times (\vec{a} \times \vec{b})]$ (D) None of these
7. Let z_1, z_2 be two complex numbers with $|z_1| = |z_2|$, then the value of $\frac{(z_1 + z_2)^2}{z_1 z_2} + (\arg(z_1) - \arg(z_2))^2$ can be
- (A) $\frac{7}{2}$ (B) 4 (C) $\frac{13}{3}$ (D) $\frac{17}{4}$
8. If $|z - (\alpha^2 - 7\alpha + 11 + i)| = 1$ and principal $\arg(z) \geq \frac{\pi}{2}$ is satisfied for atleast one z , then
- (A) Number of integral values of α is 4 (B) Number of integral values of α is 3
- (C) Exhaustive set of α lies in $[3, 8]$ (D) Exhaustive set of α lies in $[1, 7]$
9. Consider a plane P_1 passing through point $(1, 1, 1)$ and perpendicular to vector $\hat{i} + \hat{j} + \hat{k}$. If $A(1, 2, 3)$, $B(3, 1, 2)$ are any two points in spaces, a point D on plane P_1 such that $|AD - BD|$ is minimum and a plane P_2 passing through A, B and D , then
- (A) Equation of plane P_2 is $y - z + 1 = 0$
- (B) Point D is $\left(1, \frac{1}{2}, \frac{3}{2}\right)$
- (C) Line of intersection of P_1 and P_2 is $\frac{x}{-2} = y - 1 = z - 2$
- (D) Image of P_2 in P_1 does not exist
10. Let $\vec{a}, \vec{b}, \vec{c}$ be three coplanar unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$. If three vectors $\vec{p}, \vec{q}, \vec{r}$ parallel to $\vec{a}, \vec{b}, \vec{c}$ respectively and having integral but different magnitudes, then among the following options $|\vec{p} + \vec{q} + \vec{r}|$ can take a value equal to
- (A) $\frac{8}{3}$ (B) $\frac{8}{5}$ (C) $\frac{7}{4}$ (D) $\sqrt{\frac{11}{3}}$
11. Each of the circles $|z - 1 + i| = 1$ and $|z - 1 - i| = 1$ touches internally a circle of radius 2. The equation of the circle touching all the three circles can be
- (A) $3z\bar{z} + z + \bar{z} - 1 = 0$ (B) $3z\bar{z} - 7(z + \bar{z}) + 15 = 0$
- (C) $z\bar{z} - z - \bar{z} - 3 = 0$ (D) $3z\bar{z} + (z + \bar{z}) - 1 = 0$

12. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $6|\vec{a}| = 3|\vec{b}| = 2|\vec{c}| = 6$ and $|\vec{a} - 2\vec{b} + \vec{c}|^2 + |\vec{a} + \vec{b} - 2\vec{c}|^2 + |-2\vec{a} + \vec{b} + \vec{c}|^2 = 126$, then
- (A) $\vec{a}, \vec{b}, \vec{c}$ are coplanar (B) $\vec{a}, \vec{b}, \vec{c}$ constitute a triangle
(C) angle between vectors $\vec{a}$ and $\vec{b}$ is 0 (D) $\vec{a} \times ((\vec{a} \times \vec{b}) \times (\vec{a} \times (\vec{a} \times \vec{b}))) = 0$
13. During weekends, 'A' delivers newspaper in the complex plane. On Saturday, he being at z and delivers newspaper to houses located at $z^3, z^5, z^7, \dots, z^{2017}$ in that order. On Sunday, he being at 1 and delivers newspaper to houses located at $z^2, z^4, z^6, \dots, z^{2016}$ in that order. Mr. A travels in a straight line between two houses. The distance he must travel from his starting point to the last house is $\sqrt{2016}$ on each of the two days, then
- (A) $|z| = 1$ (B) $|z| = \frac{1}{2}$ (C) $\operatorname{Re}(z^2) = \frac{1008}{1009}$ (D) $\operatorname{Re}(z^2) = \frac{1007}{1008}$
14. Let P_1 be a plane containing three points $A(3, 2, 7)$, $B(6, 2, 0)$ and $C(-2, 0, 4)$. P_2 is another plane such that points B & C are mirror images of each other in this plane P_2 . Identify the correct statement(s).
- (A) Line of intersection of planes P_1 and P_2 is $\frac{26x - 11}{41} = y = \frac{26z + 43}{95}$
(B) Plane P_2 intersects the side AC of $\triangle ABC$
(C) Plane P_2 intersects the side AB of $\triangle ABC$
(D) Perpendicular distance of the plane P_2 from A is less than its perpendicular distance from B .
15. z is a uni-modular complex number such that $\operatorname{Re}(z - 1) + \operatorname{Re}(z^2) = \int_0^{\pi/2} \sin x \cdot \ln |\sin x - \cos x| dx$, if
- (A) $z + \bar{z} = -2$ (B) $z + \bar{z} = 1$ (C) $\arg(z) = \frac{\pi}{3}$ (D) $\arg(z) = \pi$
16. The position vector of the vertices A, B & C of a tetrahedron $ABCD$ are $(1, 1, 1)$, $(1, 0, 0)$ & $(3, 0, 0)$ respectively. The altitude from the vertex D to the opposite face ABC meets the median through A of $\triangle ABC$ at point E . If $AD = 4$ units and volume of tetrahedron $= \frac{2\sqrt{2}}{3}$, then
- (A) Length of altitude from vertex D is 2 units (B) There is only one possible position for point E
(C) There are two possible positions for point E (D) vector $\hat{j} - \hat{k}$ is normal to the plane ABC
17. Let z_1, z_2, z_3 be three unequal & non zero complex numbers represent vertices of $\triangle ABC$ and $|z_1 - z_2| = a$, $|z_2 - z_3| = b$, $|z_3 - z_1| = c$. If the matrix $P = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ is singular and matrix $Q = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$, then
- (A) Area of circumcircle of $\triangle ABC$ is twice the area of incircle of $\triangle ABC$.
(B) Area of circumcircle of $\triangle ABC$ is four times the area of incircle of $\triangle ABC$.
(C) $\operatorname{adj}(\operatorname{adj} Q) = (abc)^4$
(D) $\operatorname{adj}(\operatorname{adj} Q)$ is a null matrix

18. If $(1 + x + x^2 + x^3)^2 = x^3$ has roots $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6$ where $\alpha_r = \lambda_r (\cos(2\pi p_r) + i \sin(2\pi p_r))$, $0 < p_1 < p_2 < p_3 < p_4 < p_5 < p_6 < 1$, $r = 1, 2, 3, 4, 5, 6$ & λ_r is a positive real number, then
- (A) $\alpha_1 + \overline{\alpha_1} = \frac{\sqrt{5}-1}{2}$ (B) $\alpha_2 + \overline{\alpha_2} + 1 = 0$ (C) $\sum_{r=1}^6 p_r = 6$ (D) $\frac{\alpha_5 + \overline{\alpha_5}}{\alpha_4 + \overline{\alpha_4}} = \frac{\sqrt{5}-1}{2}$
19. Let two mutually perpendicular lines which are concurrent with the line $L : \frac{x-1}{-1} = \frac{y+2}{2} = \frac{z-4}{0}$ at $A \equiv (-1, 2, 4)$ and meet the plane $P : 2x - y + z = 4$ at B and C . If ΔABC is isosceles triangle and line L meets the plane P at D , then
- (A) area of $\Delta ABC = \frac{4}{3}$ sq. units. (B) area of $\Delta ABC = \frac{8}{3}$ sq. units.
(C) $AD = \sqrt{5}$ units. (D) $AD = \sqrt{6}$ units.
20. Consider three non collinear points A, B, C with their position vectors $\vec{a}, \vec{b}, \vec{a} - \vec{b}$; w.r.t. origin 'O'. 'O' is not coinciding with any of the points A, B or C . L_1 is the straight line passing through the point A and is parallel to $\vec{b}$. L_2 is the straight line passing through the point B and is parallel to $\vec{a}$. If L_3 is the straight line passing through the point C and the point of intersection of the lines L_1 and L_2 , then
- (A) the perpendicular distance of a point $D((m+1)(\vec{a} + \vec{b}))$; ($m > 0$) from the line L_3 is $m|(\vec{a} \cdot \hat{b})\hat{b} - \vec{a}|$
(B) the perpendicular distance of a point $D((m+1)(\vec{a} + \vec{b}))$; ($m > 0$) from the line L_3 is $(m+1)|(\vec{a} \cdot \hat{b})\hat{b} - \vec{a}|$
(C) the foot of perpendicular from D on the line L_3 is $\vec{a} + \left\{ (m+1) + m \frac{(\vec{a} \cdot \vec{b})}{\vec{b} \cdot \vec{b}} \right\} \vec{b}$
(D) the foot of perpendicular from D on the line L_3 is $\vec{a} + \left\{ m + (m+1) \frac{(\vec{a} \cdot \vec{b})}{\vec{b} \cdot \vec{b}} \right\} \vec{b}$
21. Let a, b, c be distinct complex numbers with $|a| = |b| = |c| = 1$ and z_1, z_2 represented by P & Q respectively are the roots of the equation $az^2 + bz + c = 0$ with $|z_1| = 1$ and $\angle POQ = \theta$; O being origin, then
- (A) $b^2 = ac$ (B) $\theta = \frac{2\pi}{3}$ (C) $PQ = \sqrt{3}$ (D) $|z_1 + z_2| = 1$
22. Let z_1, z_2, z_3 be 3 distinct complex numbers such that $\frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0$, $|z_1| = |z_2| = |z_3|$ and $\frac{z_1 \bar{z}_2 + z_2 \bar{z}_3 + z_3 \bar{z}_1 - \bar{z}_1 z_2 - \bar{z}_2 z_3 - \bar{z}_3 z_1}{(z_1 - z_2)(\overline{z_1 - z_2})} = k$, then
- (A) k^2 is greater than $\sqrt{7}$ (B) k^2 is less than $\sqrt{7}$
(C) k^4 lies in $(6, 11)$ (D) k^4 lies in $(10, 17)$

23. If the plane $ax + by = 0$ is rotated through an angle ' α ' about its line of intersection with the plane $z = 0$, then the equation of plane in new position is
- (A) $ax + by + \left(\sqrt{a^2 + b^2} \tan \alpha\right) z = 0$ (B) $ax + by + \left(\sqrt{a^2 + b^2} \cot \alpha\right) z = 0$
- (C) $ax + by - \left(\sqrt{a^2 + b^2} \cot \alpha\right) z = 0$ (D) $ax + by - \left(\sqrt{a^2 + b^2} \tan \alpha\right) z = 0$
24. Consider two planes $P_1 : 2x - 3y + z = -5$ and $P_2 : x - 4y - 14z = 5$. If a line L which is perpendicular to the line of intersection of planes P_1 & P_2 and whose one point $A(-7, y_1, z_1)$ lies on the plane P_1 and other points $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ lie on plane P_2 has the equation $\frac{x+7}{a} = \frac{y-y_1}{-b} = \frac{z-z_1}{c}$, where $a, b, c \in \mathbb{N}$, then (B & C are distinct points)
- (A) least value of $(a + b + c)$ is 6 (B) $y_1 + z_1$ is equal to -3
- (C) $\frac{z_3 - z_2}{y_3 - y_2} = \frac{1}{3}$ (D) $\frac{x_3 - x_2}{z_3 - z_2} = 2$
25. Let $z_r; r = 1, 2, 3, \dots, n$ are n distinct roots of the equation ${}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n = 0$ in argand plane. If there exist exactly one $z_r, r \in \{1, 2, 3, \dots, n\}$ such that $\arg \frac{(z_r - (-1 + \sqrt{2}i))}{((-1) - (-1 + \sqrt{2}i))} = \frac{\pi}{4}$, then n can be
- (A) 5 (B) 8 (C) 12 (D) 16
26. Let z_1, z_2 and z_3 be three distinct complex numbers satisfying $|z_1| = |z_2| = |z_3| = 1$, then which of the following is/are **CORRECT**?
- (A) $|z_1 z_2 + z_2 z_3 + z_3 z_1| = |z_1 + z_2 + z_3|$
- (B) If $|z_1 - z_2| = \sqrt{2} |z_1 - z_3| = \sqrt{2} |z_2 - z_3|$, then $\operatorname{Re} \left(\frac{z_3 - z_1}{z_3 - z_2} \right) = 0$
- (C) If $|z| > 1$ and $\arg \left(\frac{z_1}{z_2} \right) = \frac{\pi}{2}$, then $\arg \left(\frac{z - z_1}{z - z_2} \right) > \frac{\pi}{4}$
- (D) $\operatorname{Im} \left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 z_2 z_3} \right) = 0$
27. **Match the column**
- Let ABCDEF is a regular hexagon $A(z_1), B(z_2), C(z_3), D(z_4), E(z_5), F(z_6)$ in argand plane, where A, B, C, D, E and F are taken in anticlockwise sense with $z_1 = -2$ & $z_3 = 1 - \sqrt{3}i$.
- | Column-I | Column-II |
|---|-----------|
| (A) If $z_2 = a + ib$, then $2a^2 + b^2$ is equal to | (p) 8 |
| (B) The square of the inradius of hexagon is | (q) 7 |
| (C) The area of region formed by point $P(z)$ lying inside the incircle of hexagon and satisfying $\frac{\pi}{3} \leq \arg(z) \leq \frac{5\pi}{6}$ is $\frac{m}{n} \pi$, where m, n are relatively prime, then $m + n$ is equal to | (r) 5 |
| (D) The value of $z_4^2 - z_1^2 - z_2^2 - z_3^2 - z_5^2 - z_6^2$ is equal to | (s) 3 |

28. Match the column

A(2, -1, 4), B(1, 0, -1), C(1, 2, 3), D(a, b, c) be the vertices of a parallelogram ABCD taken in order. Let π be the plane containing parallelogram ABCD. Let L be the line passing through P(2, 1, 0) and parallel to vector $\hat{i} + \hat{j} + \hat{k}$. Let F be the foot of perpendicular from P to plane π and I is mirror image of P in plane π .

Column-I		Column-II		Column-III	
(A)	Coordinates of D : (2, 1, 8)	(P)	Equation of line AB $\vec{r} = (1, 0, -1) + \delta(1, -1, 5);$ $\delta \in \mathbb{R}$	(X)	Distance between AB and CD is $2\sqrt{2}$
(B)	Coordinates of $F : \left(\frac{13}{27}, \frac{19}{27}, \frac{4}{27}\right)$	(Q)	Equation of line PF $\vec{r} = (-5, -1, 1) + \delta(7, 2, -1);$ $\delta \in \mathbb{R}$	(Y)	Acute angle between line L and plane π is $\cos^{-1}\left(\frac{8}{9\sqrt{2}}\right)$
(C)	Coordinates of $I : \left(\frac{-2}{27}, \frac{11}{27}, \frac{8}{27}\right)$	(R)	Equation of projection of line L on plane π $\vec{r} = (1, 0, -1) + \delta(1, -19, -31);$ $\delta \in \mathbb{R}$	(Z)	Equation of plane, containing line L and perpendicular to plane π is $3x - 8y + 5z + 3 = 0$
(D)	Foot of perpendicular from A to BC : $\left(2, \frac{9}{5}, \frac{13}{5}\right)$	(S)	Equation of mirror image of line L in plane π $\vec{r} = (1, 0, -1) + \delta(29, -11, -35);$ $\delta \in \mathbb{R}$	(W)	Equation of plane, passing through P and parallel to plane π is $7x + 2y - z - 16 = 0$

29. Let $z_0 = \cos\left(\frac{2\pi}{2n+1}\right) + i \sin\left(\frac{2\pi}{2n+1}\right)$, $n \geq 0$ and $z = \frac{1}{2} + z_0 + z_0^2 + \dots + z_0^n$, then $(2z + 1)^{2n+1} + (2z - 1)^{2n+1}$ is equal to
30. Let $z_0, z_1, z_2, \dots, z_n$ where $n \in \mathbb{N}$ be 'n' complex numbers where z_0 be non-real cube root of 8 such that $(k+1)z_{k+1} - \omega(n-k)z_k = 0$ for all $k \in \{0, 1, 2, \dots, (n-1)\}$ where ω be non-real cube root of unity, then the value of $|z_0 + z_1 + z_2 + \dots + z_n|$ is
31. A variable line passing through the point P(0, 0, 2) always makes an angle of 60° with z-axis, intersecting the plane $x + y + z = 1$. If the locus of point of intersection of line and plane is $x^2 + y^2 = k(z - 2)^2$, then k is equal to
32. Let line L : $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-2}{0}$ meets the plane $\Pi_1 : 2x + y + z = 9$ at P and plane $\Pi_2 : y = 0$ at Q, and image of Q in the plane Π_1 is R. If the area of triangle PQR, is equal to $\frac{p}{q}\sqrt{21}$, then the value of $(p - q)$ equals
33. Let $P(z) = z^3 + az^2 + bz + c$ where a, b and c are real. There exists a complex number ω such that the three roots of $P(z)$ are $\omega + 3i$, $\omega + 9i$ and $2\omega - 4$ where $i^2 = -1$, then the value of $|a + b + c|$ is

34. The length of an edge of a cube $ABCD A_1 B_1 C_1 D_1$ is equal to unity. A point E taken on the edge $\overrightarrow{AA_1}$ is such that $\left(\frac{\overrightarrow{AE}}{\overrightarrow{AA_1}}\right) = \frac{1}{3}$ and a point F is taken on the edge $\overrightarrow{BC}$ such that $\left(\frac{\overrightarrow{BF}}{\overrightarrow{BC}}\right) = \frac{1}{4}$. If O is the centre of cube such that shortest distance of vertex B_1 from the plane of the ΔOEF is $\frac{\lambda}{\sqrt{170}}$, then λ is
35. If $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ are four concyclic points in Argand's plane and $2z_1 + 3z_2 = 4z_3 + z_4$, then the value of $\frac{|z_3 - z_4|^2}{|z_1 - z_2|^2}$ is
36. Let P be the plane containing the line $y + z = 2$, $x = 0$ and parallel to the line $x - z = 2$, $y = 0$. If plane P meets co-ordinate axes at A, B and C, then one-hundredth of the square of area of triangle ABC is
37. Let z be a complex number such that $\left|2z + \frac{1}{z}\right| = 1$ and $\arg z = \theta$, then the minimum value of $8 \sin^2 \theta$ is
38. Consider a complex number z on the argand plane satisfying $\arg(z^2 - \omega^2) = \frac{\pi}{2} + \arg(z^2 - \omega)$ (where $\omega = e^{\frac{i2\pi}{3}}$). If the minimum value of $|z - 2 - 2i| |z + 2 + 2i|$ is $\left(\frac{\sqrt{a} - \sqrt{b}}{2}\right)$ ($a, b \in \mathbb{N}$), then the reciprocal of the square root of the greatest integer of the square root of $(a + b)$ is
39. If 'a' is real constant greater than 2 and A, B, C are variable angles satisfying $\sqrt{a^2 - 4} \tan A + a \tan B + \sqrt{a^2 + 4} \tan C = 6a$, then the least value of $\tan^2 A + \tan^2 B + \tan^2 C$ is
40. Let $\vec{a} = x^2 \hat{i} - 3\hat{j} + (x+3)\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - (x-3)\hat{k}$ be two vectors such that $|\vec{a}| = |\vec{b}|$. If $\vec{v}_1 = 2\vec{a} + 3\vec{b}$ and $\vec{v}_2 = 3\vec{a} - 2\vec{b}$ and $I = \int_1^3 (\vec{v}_1 \wedge \vec{v}_2) dx$, then the value of $[I]$ is
[Note: $[k]$ denotes greatest integer function less than or equal to k .]
41. If z_1, z_2, z_3 be equimodular non zero complex numbers such that $z_1 + z_2 + z_3 = 0$, then the value of $\operatorname{Re}\left(\sum_{j=1}^3 \sum_{k=1}^3 \frac{z_j}{z_k}\right)$ is
42. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{\ell}, \vec{m}, \vec{n}$ are two sets of unit vectors such that $[\vec{a} \vec{b} \vec{c}] = 1$ and $\vec{\ell}$ makes equal angles of 60° with vectors $\vec{a}$ and $\vec{b}$, $\vec{m}$ is equally inclined to $\vec{b}$ and $\vec{c}$ at an angle of 45° and $\vec{n}$ is equally inclined to all three vectors $\vec{a}, \vec{b}, \vec{c}$, such that the reciprocal of volume of parallelopiped formed by vectors $\vec{\ell}, \vec{m}, \vec{n}$ is $\sqrt{a}(\sqrt{b} + 1)$, then $\frac{2}{2a + b}$ is
43. If $\vec{V}_1 = \hat{i} + \hat{j} + \hat{k}$; $\vec{V}_2 = a\hat{i} + b\hat{j} + c\hat{k}$ where $a, b, c \in \{-2, -1, 0, 1, 2\}$, then the number of non zero vectors $\vec{V}_2$ which are perpendicular to $\vec{V}_1$ is

ANSWER KEY

- | | | | | | | | | | | | | | | |
|-----|--------|-----|--|-----|--------|-----|-------|-----|-------|-----|-------|----------------|--------|--|
| 1. | (A) | 2. | (C) | 3. | (A) | 4. | (C) | 5. | (B) | 6. | (C) | 7. | (BCD) | |
| 8. | (AD) | 9. | (ABCD) | | | 10. | (ACD) | 11. | (AB) | 12. | (ACD) | 13. | (AD) | |
| 14. | (ACD) | 15. | (ABCD) | | | 16. | (ACD) | 17. | (BD) | 18. | (ABD) | 19. | (BC) | |
| 20. | (AC) | 21. | (ABCD) | | | 22. | (BC) | 23. | (AD) | 24. | (ABD) | 25. | (BD) | |
| 26. | (ABD) | 27. | (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (q); (D) $\rightarrow$ (p) | | | | | | | | 28. | (AC, PQRS, XW) | | |
| 29. | (0) | 30. | (2) | 31. | (3) | 32. | (7) | 33. | (136) | 34. | (14) | 35. | (1.50) | |
| 36. | (0.12) | 37. | (7) | 38. | (0.25) | 39. | (12) | 40. | (3) | 41. | (0) | 42. | (0.04) | |
| 43. | (18) | | | | | | | | | | | | | |